

Math 110 – Homework 3 Answers

Math 110 – Fall 2026

Homework 3

Linear maps, null spaces, ranges, and matrices

Suggested completion date: Friday, September 18

Unless otherwise indicated, all vector spaces are over the field $\mathbb{F}$. Problems referring to LADR are from *Linear Algebra Done Right*, 4th ed..

Problem 1. Let $T : V \rightarrow W$ be linear, with V finite-dimensional, and let U be a subspace of W .

- (a) Prove that $T^{-1}(U) = \{v \in V : Tv \in U\}$ is a subspace of V .
- (b) Prove the dimension formula

$$\dim T^{-1}(U) = \dim \text{null } T + \dim(U \cap \text{range } T).$$

- (c) Let $D : \mathcal{P}_4(\mathbb{R}) \rightarrow \mathcal{P}_4(\mathbb{R})$ be differentiation, $Dp = p'$. Determine, using part (b), the dimension of the space of polynomials p for which

$$p'(x) = ax^2 + b$$

for some $a, b \in \mathbb{R}$.

Problem 2.

- (a) Find a linear map $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ such that $\text{null } T = \text{range } T$.
- (b) Prove that no linear map $T : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ can satisfy $\text{null } T = \text{range } T$.

Problem 3. Return to the subspace from Homework 2,

$$U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}.$$

Define a suitable linear map $S : \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^2$ having U as its null space, and use rank-nullity to determine $\dim U$. Compare this calculation with the direct basis calculation from Homework 2.

Problem 4. Let $T : V \rightarrow W$ be linear.

- (a) Prove that if $v_1, \dots, v_n$ is linearly dependent, then $Tv_1, \dots, Tv_n$ is linearly dependent.
- (b) Give an example showing that a linearly independent list need not be sent to a linearly independent list.

Problem 5. Define $T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathcal{P}_4(\mathbb{R})$ by

$$(Tp)(x) = x^2 p'(x) + p(x).$$

- (a) Find the matrix of T with respect to the standard monomial bases

$$1, x, x^2, x^3 \quad \text{and} \quad 1, x, x^2, x^3, x^4.$$

- (b) Determine $\text{null } T$ and $\text{range } T$, and give their dimensions.

- (c) Explain directly from your answer to (a) why the range of T is the span of the columns of its representing matrix.

Problem 6. Solve the system

$$\begin{aligned}x + y + z + w &= 1, \\x + z &= 0\end{aligned}$$

over the field $\mathbb{F}_2$. List all solutions.

Problem 7 (optional). Let A be an $n \times m$ matrix over $\mathbb{F}_2$, and suppose $Ax = b$ is consistent. Express the number of solutions in terms of m and $\text{rank } A$.

Solutions

Problem 1

Solution notes

Problem 2

Solution notes

Problem 3

Solution notes

Problem 4

Solution notes

Problem 5

Solution notes

Problem 6

Solution notes

Problem 7

Solution notes
